

CSA17 – Artificial Intelligence

Assessment Tool 1 – Analytical Problem Solving: Complete Solution

Question 1 – Two-Room Vacuum World

The vacuum world has two squares A and B. Initially both are dirty. The agent starts at A. The performance measure gives +1 for every clean square at the end of each step, with a penalty p for every Left/Right movement.

(a) PEAS Description

PEAS	Description
P – Performance	+1 for every clean square at the end of each time step; $-p$ for every Left/Right movement
E – Environment	Two squares A and B containing dirt; dirt does not reappear
A – Actuators	Suck, Left, Right, NoOp
S – Sensors	Detects current location and whether the square is Dirty/Clean

(b) Agent X – Simple Reflex

- If current square is dirty $\rightarrow$ Suck.
- Otherwise $\rightarrow$ move to the other square.

Step	Location	Percept	Action	Clean squares	Move penalty	Step score
1	A	Dirty	Suck	A	0	1
2	A	Clean	Right	A	1	0
3	B	Dirty	Suck	A, B	0	2
4	B	Clean	Left	A, B	1	1
5	A	Clean	Right	A, B	1	1
6	B	Clean	Left	A, B	1	1

Clean-square score = 10. Agent X makes 4 movements. For $p = 1$: Total score = $10 - 4(1) = 6$.

Answer: Agent X total score = 6.

(c) Agent Y – Reflex with Internal State

Agent Y remembers whether both squares have already been cleaned. After cleaning both squares, it performs NoOp instead of moving unnecessarily.

Step	Location	Percept	Action	Clean squares	Move penalty	Step score
1	A	Dirty	Suck	A	0	1
2	A	Clean	Right	A	1	0
3	B	Dirty	Suck	A, B	0	2

4	B	Clean	NoOp	A, B	0	2
5	B	Clean	NoOp	A, B	0	2
6	B	Clean	NoOp	A, B	0	2

Total score = $1 + 0 + 2 + 2 + 2 + 2 = 9$.

Answer: Agent Y total score = 9.

(d) Score as a Function of p

Agent X makes 4 movements: $\text{Score}(X) = 10 - 4p$.

Agent Y makes 1 movement: $\text{Score}(Y) = 10 - p$.

Y is strictly better when: $10 - p > 10 - 4p \rightarrow 3p > 0 \rightarrow p > 0$.

When $p = 0$, both agents score 10.

(e) Which Agent is Rational?

Agent Y is rational under this performance measure. It remembers that both squares have already been cleaned and therefore chooses NoOp rather than repeatedly moving between clean squares. Agent X continues moving because it does not have internal memory. Rationality depends on the performance measure and the information available from the percept sequence.

Question 2 – Task Environment Analysis

Environments: E1 – Autonomous taxi; E2 – Part-picking robot; E3 – Chess-playing agent. The harder property receives 1 point and semi-dynamic receives 0.5 point.

(a) Classification

E1 – Autonomous Taxi

Property	Classification	Reason
Observability	Partially observable	Taxi cannot observe everything around it
Agents	Multi-agent	Other vehicles/drivers are involved
Determinism	Stochastic	Traffic events are uncertain
Episodic/Sequential	Sequential	Current decisions affect later driving
Static/Dynamic	Dynamic	Traffic and surroundings change
Discrete/Continuous	Continuous	Position, speed and movement vary continuously

Difficulty score = $1 + 1 + 1 + 1 + 1 + 1 = 6$.

E2 – Part-Picking Robot

Property	Classification	Reason
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Observability	Partially observable	Camera cannot provide perfect information
Agents	Single agent	One robot performs the task
Determinism	Stochastic	Perception/picking involves uncertainty
Episodic/Sequential	Sequential	Actions occur continuously as parts arrive
Static/Dynamic	Dynamic	Conveyor and parts are moving
Discrete/Continuous	Continuous	Physical movement is continuous

Difficulty score = $1 + 0 + 1 + 1 + 1 + 1 = 5$.

E3 – Chess Agent

Property	Classification	Reason
Observability	Fully observable	Board state is visible
Agents	Multi-agent	Two players compete
Determinism	Deterministic	A legal move has a predictable board-state result
Episodic/Sequential	Sequential	Each move affects future moves
Static/Dynamic	Semi-dynamic	Board is static, but the game clock changes
Discrete/Continuous	Discrete	Moves and board states are discrete

Difficulty score = $1 + 1 + 0 + 1 + 0.5 + 0 = 3.5$.

(b) Difficulty Ranking

Environment	Score
E1 – Autonomous Taxi	6
E2 – Part-Picking Robot	5
E3 – Chess	3.5

Hardest → E1 → E2 → E3 → Easiest.

(c) Agent Architecture

Hardest – E1 Autonomous Taxi: A utility-based agent is suitable because it can consider safety, travel time, traffic, other vehicles and passenger objectives.

Easiest – E3 Chess: A goal-based agent is suitable because it has a clear goal: win the chess game/checkmate the opponent.

(d) Interpretation

AI environment complexity depends on properties such as observability, number of agents, uncertainty, sequential decisions, dynamic changes and continuous variables. More difficult environments generally require more sophisticated agent programs.

Question 3 – Turing Test Data

The question gives 50 interrogations for each system. Correctly identified as a machine: ELIZA = 47, Chatbot B = 36, System C = 29.

(a) Correct-Identification Rate and Fooling Rate

Correct-identification rate = Correctly identified / Total $\times$ 100.

Fooling rate = (Total – Correctly identified) / Total $\times$ 100.

System	Correct identification	Fooling rate
ELIZA	94%	6%
Chatbot B	72%	28%
System C	58%	42%

(b) Turing's 70% Criterion

Turing's criterion in the question states that an average interrogator would have no more than a 70% chance of making the correct identification after five minutes. Pure guessing gives 50%.

System	Correct rate	Meets $\leq 70\%$ criterion?
ELIZA	94%	No
Chatbot B	72%	No
System C	58%	Yes

Therefore, System C satisfies the stated 70% criterion.

(c) Improvement in Fooling Rate

ELIZA $\rightarrow$ System C: $42\% - 6\% = 36$ percentage points; $42/6 = 7$, so System C's fooling rate is 7 times ELIZA's.

Chatbot B $\rightarrow$ System C: $42\% - 28\% = 14$ percentage points.

(d) Interpretation

The Turing Test is mainly related to the AI approach of acting humanly. A high fooling rate means judges were more likely to mistake the machine for a human in the text-conversation experiment. However, a high fooling rate does not by itself prove human-like thinking, understanding or rationality.

Question 4 – Rational Route Choice Under Uncertainty

Utility function: $U = 100 - T - 20L$, where T is travel time and $L = 1$ if arrival is after 55 minutes, otherwise $L = 0$.

(a) Utility of Each Route

Route	Outcome	Time	Late?	Utility
R1	Normal	30	No	70
R1	Disrupted	70	Yes	10
R2	Normal	45	No	55

R2	Disrupted	60	Yes	20
R3	Normal	50	No	50

R1 normal: $100 - 30 = 70$. R1 disrupted: $100 - 70 - 20 = 10$.

R2 normal: $100 - 45 = 55$. R2 disrupted: $100 - 60 - 20 = 20$.

R3: $100 - 50 = 50$.

(b) Expected Utility

$EU = \sum \text{Probability} \times \text{Utility}$.

R1: $(0.70 \times 70) + (0.30 \times 10) = 49 + 3 = 52$.

R2: $(0.90 \times 55) + (0.10 \times 20) = 49.5 + 2 = 51.5$.

R3: $1.00 \times 50 = 50$.

Route	Expected Utility
R1	52
R2	51.5
R3	50

Therefore, R1 is the rational choice in part (b).

(c) New Highway Accident Alert

New R1 disruption probability = 0.80, so normal probability = 0.20.

$EU(R1) = (0.20 \times 70) + (0.80 \times 10) = 14 + 8 = 22$.

R2 remains 51.5 and R3 remains 50. Therefore, R2 becomes the highest expected-utility choice.

(d) Rational vs Omniscient Agent

If the disruption actually occurred on R1, the actual utility was 10. The agent was not irrational because it did not know with certainty that the accident would occur. It used the available probabilities and selected the action with the highest expected utility.

A rational agent chooses the action expected to give the best performance based on available information. An omniscient agent would know the actual future outcome in advance. Rationality does not require perfect knowledge of the future.

(e) Interpretation

Rationality depends on the percept sequence, performance measure, available actions and expected outcomes. In part (b), R1 had the highest expected utility (52), so choosing R1 was rational. In part (c), the new radio alert changed R1's disruption probability from 30% to 80%, reducing its expected utility from 52 to 22. Therefore, the rational decision changed even though the roads themselves did not change.

Final Answers at a Glance

Question	Main Final Answer
Q1(b)	Agent X score = 6 when $p = 1$
Q1(c)	Agent Y score = 9 when $p = 1$
Q1(d)	$X = 10 - 4p$; $Y = 10 - p$; Y better when $p > 0$
Q1(e)	Agent Y is rational under the given performance measure
Q2	$E1 = 6$, $E2 = 5$, $E3 = 3.5$
Q2(b)	$E1 \rightarrow E2 \rightarrow E3$ by the given difficulty score
Q3(a)	ELIZA: 94%/6%; B: 72%/28%; C: 58%/42%
Q3(b)	System C satisfies the $\leq 70\%$ criterion
Q3(c)	ELIZA $\rightarrow$ C = 36 percentage points, $7\times$; B $\rightarrow$ C = 14 percentage points
Q4(a)	R1: 70/10; R2: 55/20; R3: 50
Q4(b)	$R1 = 52$, $R2 = 51.5$, $R3 = 50$
Q4(c)	New R1 EU = 22; R2 becomes the highest-EU choice
Q4(d)	Actual R1 utility after disruption = 10; choosing R1 was not irrational